Topic 04: Insurance Charges

Nguyen Tan Dat
FPT University
SE190212

Nguyen Khanh Duy
FPT University
SE191051

***Abstract*—This study addresses the challenge of estimating individual medical insurance charges within the field of health analytics using machine learning methodologies. Precise cost prediction remains a significant challenge due to the inherent nature of insurance data, which typically exhibits a heavy-tailed distribution and contains numerous outliers. We propose a comprehensive machine learning pipeline centered on the logarithmic transformation of the target variable. The study evaluates and compares linear models (OLS, Ridge, and Lasso) against non-linear approaches (SVR). Experimental results on the Kaggle-Insurance dataset demonstrate that log-transformation significantly improves RMSE and MAE metrics. Notably, the SVR model combined with log-transformed targets achieved the highest performance ($R^2 = 0.834$), effectively handling outliers. Furthermore, the findings confirm that smoking status and Body Mass Index (BMI) are the most sensitive features, exerting the most substantial influence on insurance costs.**

I. Introduction

In recent years, the rapid increase in healthcare costs has become a major concern for both individuals and insurance providers. Accurately estimating individual medical insurance expenses is an important problem in the field of health analytics, as it enables insurance companies to develop appropriate pricing strategies while also helping individuals make informed financial decisions. With the availability of structured datasets containing demographic and lifestyle information (e.g., age, BMI, smoking status), machine learning methods have been widely applied to model and predict insurance costs [1].

The core problem addressed in this study is to predict individual insurance charges based on personal characteristics. This is a regression task in which the target variable (charges) is typically skewed and contains many outliers. These properties make the prediction task more challenging, especially when using traditional linear models without appropriate preprocessing or transformation techniques [1].

Traditional methods such as Ordinary Least Squares (OLS) regression offer advantages in terms of simplicity and interpretability, but they are highly sensitive to outliers and rely on assumptions such as normally distributed errors [2]. Regularization techniques such as Ridge and Lasso improve model stability in the presence of multicollinearity, particularly when categorical variables are encoded into multiple features [3]. However, these linear models still struggle to capture complex relationships in the data. In contrast, nonlinear methods such as Support Vector Regression (SVR) are more flexible in modeling such relationships, but they require careful hyperparameter tuning and incur higher computational costs [4]. Furthermore, many previous studies have not systematically evaluated the impact of target variable transformation (e.g., log-transform) on model performance.

To address these limitations, this paper proposes a comprehensive machine learning pipeline for insurance cost prediction. The main contributions of this study are as follows: (1) a systematic comparison between models trained on raw data and log-transformed targets to analyze the impact of reducing distribution skewness; (2) an evaluation of multiple regression methods, including OLS, Ridge, Lasso, and SVR, under a unified experimental setting; and (3) an interpretability analysis of the models, with a particular focus on key factors such as smoking status and BMI. Through extensive experiments, this study aims to identify a model that is both robust and interpretable for practical insurance cost estimation.

- **RQ1:** Does the application of a logarithmic transformation to the target variable significantly improve $RMSE$ and MAE compared to models using raw cost data?
- **RQ2:** Do regularization techniques, specifically Ridge and Lasso regression, enhance model stability when handling high-dimensional data generated by One-hot encoding of categorical variables?
- **RQ3:** Does Support Vector Regression (SVR) demonstrate superior robustness in handling outliers compared to traditional linear models?

II. Related work

A. Linear Regression

1) Theoretical Background: Linear Regression is one of the most fundamental and widely adopted methods for regression tasks, aiming to model the linear relationship between input features and a target variable. The model operates under the assumption that the target variable y can be represented as a linear combination of the input features x [2].

2) Theoretical Background: In the case of *Simple Linear Regression* (one input variable), the relationship is defined as:

$$y = w_1 x + w_0 \tag{1}$$

For *Multiple Linear Regression* (multiple input variables), the model is extended as follows:

$$y = w_0 + w_1 x_1 + w_2 x_2 + \cdots + w_n x_n \tag{2}$$

Alternatively, it can be expressed in summation notation:

$$y = w_0 + \sum_{i=1}^{n} w_i x_i \tag{3}$$

Where:

- x: Input (independent variable).
- y: Output (dependent variable).
- w_i: Coefficients (slope or weights).
- w_0: Bias (intercept or the value of y when all $x_i = 0$).

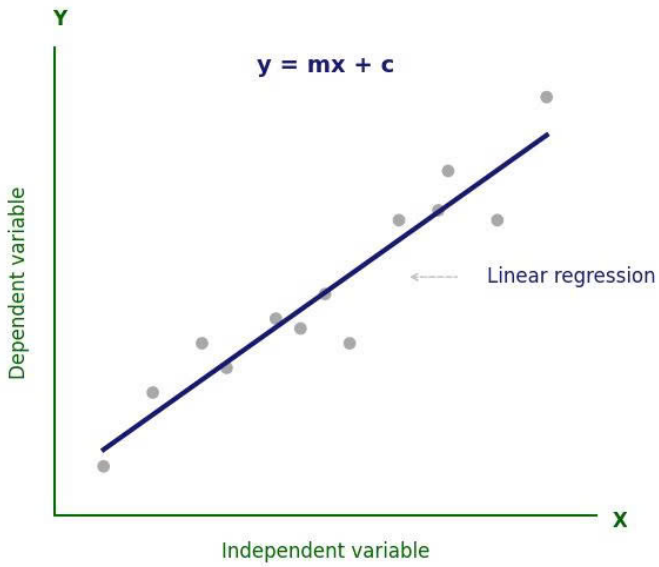

Fig. 1. Linear regression

3) Optimization Objective: The primary goal of the regression model is to determine the optimal parameters w and b that minimize the discrepancy between the predicted values and the actual observed values. This discrepancy is quantified using a loss function, specifically the Mean Squared Error (MSE), which is defined as follows:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{4}$$

Where:

- n: The total number of observations (the sample size in the insurance dataset).
- y_i: The actual insurance charges recorded in the dataset for the i-th individual.
- $\hat{y}_i$: The predicted insurance charges generated by the model (e.g., SVR, Ridge, or Lasso).
- $(y_i - \hat{y}_i)^2$: The squared error. Squaring the error ensures that all values are non-negative and serves to *heavily penalize* larger prediction errors.
- $\sum$: The summation of squared errors across all samples in the dataset.
- $\frac{1}{n}$: The averaging term, which ensures that the metric remains independent of the dataset scale.

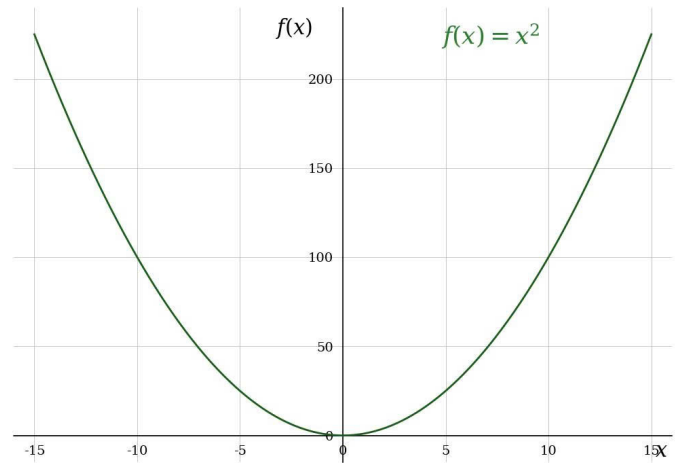

Fig. 2. Loss funtion linear regression

4) Optimization Objective: Mathematically, MSE is a **convex function**. This property is crucial as it guarantees that optimization algorithms (such as Gradient Descent or OLS) can converge to a *global optimum*, ensuring the best possible fit for the linear relationship.

Estimation Methods: The parameters of the regression model can be estimated using several approaches, most notably:

- **Ordinary Least Squares (OLS):** A non-iterative method that minimizes the sum of squared residuals.
- **Optimization Algorithms:** Such as *Gradient Descent*, which iteratively updates parameters to minimize the loss function.

For linear regression, OLS provides a **closed-form solution** (also known as the normal equation) to find the optimal weight vector $\mathbf{w}$:

$$\mathbf{w} = (X^T X)^{-1} X^T \mathbf{y} \tag{5}$$

Where:

- X: The feature matrix of size $n \times p$ (where n is the number of samples and p is the number of features).
- X^T: The transpose of the feature matrix X.
- $(X^T X)^{-1}$: The inverse of the product between the transposed matrix and the original matrix.
- $\mathbf{y}$: The vector of actual target values (insurance charges).
- $\mathbf{w}$: The vector of estimated coefficients.

5) L1 and L2 Regularization: To enhance the model's generalization capability and mitigate the risk of **overfitting**, regularization techniques are applied. These methods introduce a penalty term to the cost function to constrain the magnitude of the coefficients.

a) Ridge Regression (L2 Regularization): Ridge regression adds a penalty equivalent to the square of the magnitude of coefficients ($L2$ norm). The optimization objective is defined as:

$$\min_{w} \left[\sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{p} w_j^2 \right] \tag{6}$$

b) Lasso Regression (L1 Regularization): Lasso (Least Absolute Shrinkage and Selection Operator) adds a penalty equivalent to the absolute value of the magnitude of coefficients ($L1$ norm). The optimization objective is:

$$\min_{w} \left[\sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{p} |w_j| \right] \tag{7}$$

In which:

- λ (Lambda): The **regularization strength** (penalty parameter).
- **Lasso (L1):** Has the unique ability to shrink coefficients to **exactly zero**. By nullifying less significant predictors, Lasso inherently performs **feature selection**, resulting in a sparse and more interpretable model.

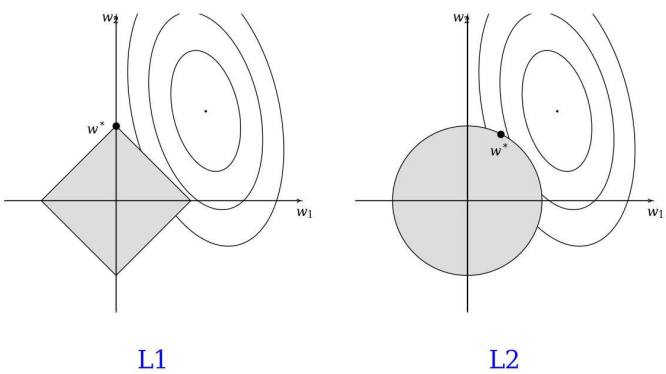

Fig. 3. Lasso and Ridge

- **Fig. Lasso Regression and Ridge Regression:** [Insert your figure reference here].
- **Ridge Regression:** This technique "smooths" the model, effectively reducing variance and fluctuations in the predictions.
- **Lasso Regression:** This method is used for feature selection by eliminating unimportant features (shrinking their coefficients to zero).

6) Advantages and Disadvantages: Understanding the strengths and limitations of linear-based models is crucial for selecting the appropriate methodology for insurance cost estimation.

a) Advantages:

- **Simplicity and Interpretability:** The models are straightforward to implement and provide highly interpretable results, allowing for a clear understanding of how each feature affects the target variable.
- **Computational Efficiency:** These methods possess low computational complexity, making them highly suitable for large-scale datasets.
- **Analytical Solution:** OLS provides a well-defined closed-form solution, ensuring a unique global optimum for the parameters.
- **Foundation for Advanced Models:** These techniques serve as the fundamental framework for more sophisticated extensions such as Ridge and Lasso regression.

b) Disadvantages:

- **Linearity Assumption:** The assumption of a linear relationship between features and the target variable may not always align with the complexities of real-world data.
- **Sensitivity to Outliers:** Linear models are highly sensitive to *outliers*, which can significantly distort the estimated coefficients and lead to biased predictions.
- **Multicollinearity:** The models are susceptible to *multicollinearity*, where highly correlated input variables can lead to unstable parameter estimates.
- **Limited Expressiveness:** They possess a limited capacity to represent and capture complex, non-linear patterns within the data.

B. Log-Transform

1) Theoretical Framework: In many empirical regression tasks, particularly in insurance charge prediction, the target variable y often exhibits a *right-skewed* and *heavy-tailed* distribution. Such characteristics violate the normality assumption of residuals in linear models, thereby diminishing predictive efficiency [2], [3].

A widely adopted technique to mitigate this issue is the **logarithmic transformation** of the target variable:

$$y' = \log(y) \tag{8}$$

Where:

- y: The original value (insurance charges).
- y': The transformed value after applying the logarithm.

After training the model on y', the predicted values can be reverted to the original scale using the exponential function (inverse transformation):

$$\hat{y} = \exp(\hat{y}') \tag{9}$$

2) Significance and Intuition: The application of Log-transform provides several key advantages:

- **Skewness Reduction:** It effectively compresses the range of values, reducing the skewness of the data.
- **Normalization:** It shifts the distribution closer to a Gaussian (Normal) distribution, which is a core assumption for many statistical estimators.
- **Outlier Mitigation:** It dampens the influence of extreme values (outliers) at the tail of the distribution.

Furthermore, a linear model fit on a log-scale can be interpreted as modeling an **exponential relationship** in the original space:

$$\log(y) = \beta x \Rightarrow y = e^{\beta x} \tag{10}$$

This allows the linear framework to capture non-linear patterns inherent in medical cost data.

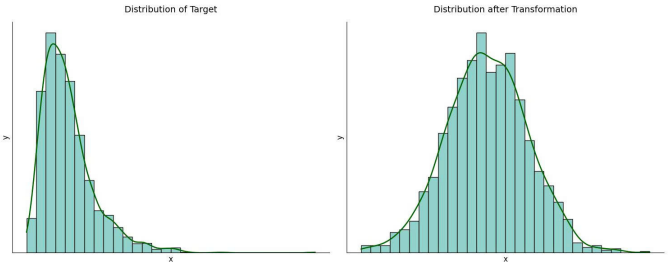

Fig. 4. Log target variable

Before log-transformation:

- Strong right-skewed distribution.
- Numerous extreme values (outliers) biasing the model.

After log-transformation:

- Distribution becomes more balanced and symmetrical.
- Data "compresses" at the tail end.
- Linear Regression achieves a better fit.

3) Optimization Objective:

4) Literature and Methodology: In health economics and financial analytics research, log-transformation is a standard procedure for handling skewed expenditure data [3]. Studies have demonstrated that applying log-transform to insurance charges significantly improves performance metrics such as $RMSE$ and MAE by minimizing the variance inflation caused by high-cost cases.

In a typical machine learning pipeline, Log-transform is often integrated with:

- **Linear Regression:** To satisfy linearity and homoscedasticity assumptions.
- **Ridge/Lasso:** To enhance stability and coefficient reliability.
- **SVR:** Although SVR is inherently robust, log-transformation can further assist in handling extreme skewness.

Current research emphasizes the importance of conducting a comparative analysis between the *raw target* and *log-transformed target* to evaluate the precise gain in model performance.

C. Support Vector Regression (SVR)

1) Theoretical Framework: Support Vector Regression (SVR), an extension of Support Vector Machines (SVM) for regression tasks, aims to find an approximation function where the errors remain within a predefined threshold ϵ, while simultaneously ensuring minimal model complexity [5].

The core intuition of SVR is to find a hyperplane that:

- Fits the data within an ϵ-insensitive zone (the "tube").
- Maximizes the margin between the hyperplane and the boundary points, known as **Support Vectors**.

2) Mathematical Representation: The prediction function (Regression Hyperplane) is defined as:

$$f(x) = \langle w, x \rangle + b \tag{11}$$

Where:

- $\langle w, x \rangle$: The dot product of the weight vector w and the input features x (e.g., age, BMI, smoking status).
- b: The bias term (intercept) that shifts the prediction line.

a) Optimization Objective: SVR constructs an "ϵ-tube" around the data points with a width of ϵ. Points falling outside this tube are penalized using *slack variables* (ξ, ξ^*). The objective is to minimize:

$$\min_{w,b,\xi,\xi^*} \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}(\xi_i + \xi_i^*) \tag{12}$$

b) Constraints: The model is subject to the following constraints for each sample i:

$$\begin{cases} y_i - \langle w, x_i \rangle - b \leq \epsilon + \xi_i \\ \langle w, x_i \rangle + b - y_i \leq \epsilon + \xi_i^* \\ \xi_i, \xi_i^* \geq 0 \end{cases} \tag{13}$$

Where:

- ϵ (Epsilon): The width of the error tube. Errors smaller than ϵ are ignored in the loss function (the "insensitive" zone).
- y_i: The actual insurance charges.
- $f(x_i)$: The predicted value.
- C: The regularization parameter that balances the flatness of the model against the amount of allowed errors outside the tube.

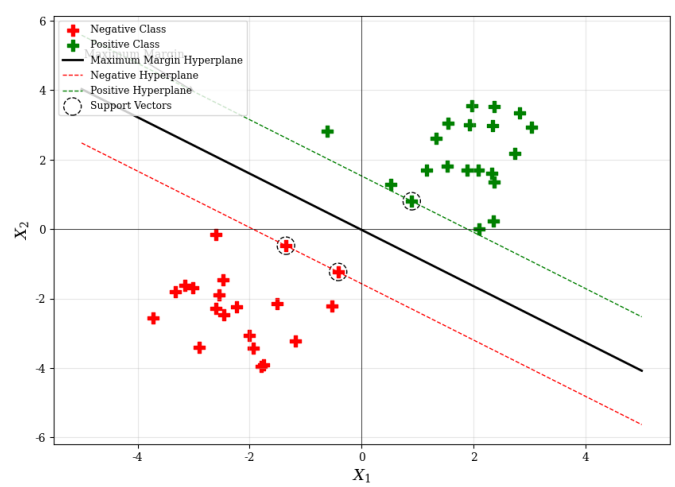

Fig. 5. SVM-Maximum Margin Classifier Visualization

3) Optimization Objective:

4) Advantages and Disadvantages:

a) Advantages:

- **High-Dimensional Efficiency:** Highly effective in spaces with many features.
- **Robustness to Overfitting:** Since the model only depends on support vectors, it is less susceptible to noise and outliers.
- **Kernel Trick:** Capable of capturing non-linear relationships by mapping data into higher-dimensional spaces using kernels (e.g., RBF, Polynomial) [6].

b) Disadvantages:

- **Computational Cost:** Performance degrades on very large datasets due to high training time.
- **Parameter Sensitivity:** Difficult to tune optimal values for the kernel, C, and ϵ.
- **Interpretability:** Less intuitive and harder to explain compared to simple Linear Regression.

D. Conclusion and Comparative Summary

The regression methodologies discussed for insurance cost prediction involve critical trade-offs between predictive accuracy, interpretability, and model stability. The selection of an optimal model depends heavily on the data characteristics (e.g., skewness, presence of outliers) and the specific objectives of the study.

- **Accuracy vs. Interpretability:** Linear models such as OLS, Ridge, and Lasso offer high interpretability, allowing for a direct analysis of how features like BMI or smoking status influence costs. However, they may underperform when the data contains complex non-linear relationships. Conversely, SVR (particularly with an RBF kernel) can achieve superior accuracy but functions more as a "black box," making its internal logic harder to explain.
- **Bias–Variance Trade-off:** OLS typically aims for low bias but can suffer from high variance, making it sensitive to noise. Regularization techniques (Ridge and Lasso) reduce variance by introducing a small amount of bias, leading to better generalization. SVR manages this trade-off via the parameter C; a high C may lead to overfitting (low bias, high variance), while a lower C encourages a smoother, more generalized model.
- **Robustness vs. Sensitivity to Outliers:** Linear regression is highly sensitive to outliers, which can significantly skew the regression line. In contrast, SVR is inherently more *robust* due to its ϵ-insensitive loss function, which ignores minor errors. Log-transformation serves as a strategic intermediary, mitigating outlier impact without requiring a transition to a more complex model.
- **Linear Simplicity vs. Non-linear Flexibility:** Linear models are computationally efficient and easy to deploy but are strictly limited to linear boundaries. SVR, through the *kernel trick*, provides the flexibility to capture intricate non-linear patterns at the cost of higher computational requirements and more intensive hyperparameter tuning.
- **Log-Transform (Performance Gain vs. Interpretability):** Applying a log-transformation significantly enhances model performance by reducing skewness and stabilizing variance. However, this comes at the cost of direct interpretability, as the results must be back-transformed to be understood on the original monetary scale.

In summary, while **Linear Regression** focuses on fitting a continuous trend line, **SVR** seeks the optimal boundary (hyperplane) that maximizes the margin. This study will evaluate these trade-offs by comparing a standard baseline against a log-transformed pipeline and a non-linear SVR approach to identify the most effective solution for the insurance dataset.

III. Main Contribution

A. Problem Foundation

This research addresses the core regression challenge of predicting individual medical insurance charges. The ability to precisely estimate these costs based on demographic information and lifestyle factors is a critical component of modern health analytics and actuarial science.

The objective is to develop a predictive model that learns the underlying patterns from a set of input features $\mathbf{X}$ to estimate the target variable y (insurance charges). The dataset utilized in this study comprises the following key features:

- **Demographic factors:** Age, Sex, and Region of residence.
- **Health and Lifestyle indicators:** Body Mass Index (BMI) and Smoking status (Smoker).
- **Family structure:** Number of children/dependents.

By formulating this as a supervised learning task, we aim to minimize the discrepancy between the predicted charges $\hat{y}$ and the actual historical charges y, thereby providing a robust tool for financial risk assessment in healthcare.

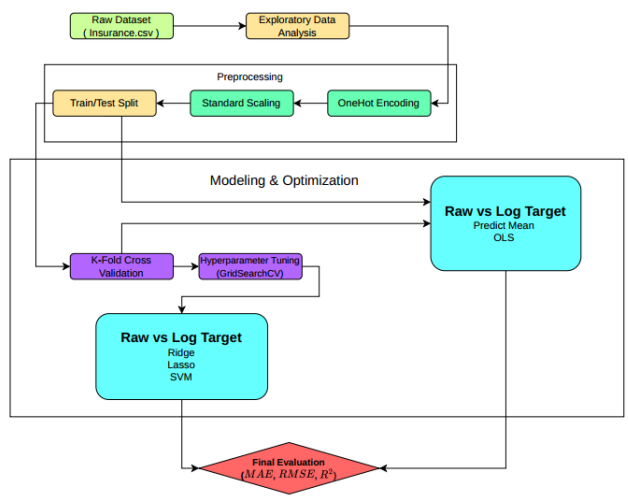

Fig. 6. Workflow

The proposed end-to-end workflow is systematically structured to ensure that each stage—from initial data ingestion to the final assessment of predictive accuracy—is executed with high precision and transparency, facilitating a robust comparison between different regression methodologies.

B. Data Engineering

The data engineering and preprocessing workflow is systematically designed to enhance the quality of information provided to the models:

- **Target Transformation (Log Transformation):** Visual inspection reveals that the target variable *charges* exhibits a significant right-skewed distribution. Consequently,

a logarithmic transformation ($np.log1p$) is applied to smooth the data, shifting the distribution closer to a Gaussian (normal) distribution and enhancing the stability of the learning algorithms.

- **Preprocessing Pipeline:** Continuous numerical variables (*age, bmi, children*) are scaled using `StandardScaler` to eliminate unit discrepancies. Categorical variables (*sex, smoker, region*) are encoded using `OneHotEncoder` with the `drop="first"` configuration to prevent multicollinearity.
- **Dataset Splitting:** The dataset is partitioned into an 80% training set and a 20% testing set using the `train_test_split` function to ensure an unbiased final evaluation.

C. Model Design

This study implements and compares various machine learning algorithms, ranging from basic linear models to more sophisticated regression techniques:

- **Training Algorithms:** Baseline (Predict Mean), OLS (Ordinary Least Squares), Ridge Regression, Lasso Regression, and SVR (Support Vector Regressor).
- **Hyperparameter Tuning:** Optimal hyperparameter grids (e.g., α for Ridge/Lasso; C, γ, ϵ, and *kernel* for SVR) are systematically screened and evaluated via `GridSearchCV`.
- **Data Leakage Prevention:** To ensure validity, `GridSearchCV` is strictly applied only to the training set. Furthermore, a `TransformedTargetRegressor` is designed to automate the log-transformation and its inverse transformation ($np.expm1$) during model execution, maintaining seamless scale consistency.

D. Evaluation Metrics

The performance of the regression models is quantified using three primary metrics:

- **MAE (Mean Absolute Error):** Measures the average absolute magnitude of errors between predictions and actual costs.
- **RMSE (Root Mean Squared Error):** The square root of the average squared discrepancies, which heavily penalizes larger prediction deviations.
- **$\mathbf{R^2}$ Score (Coefficient of Determination):** Represents the proportion of variance in the target variable that is predictable from the input features.

E. Experimental Strategy

Two parallel experimental scenarios are established to verify the effectiveness of data transformation:

- **Scenario 1:** Training and evaluating models on the original, unprocessed target variable (*Raw Target*).
- **Scenario 2:** Training and evaluating models on the log-transformed target variable (*Log Target*).

In both scenarios, K-Fold Cross-Validation ($K = 5$) is employed to estimate algorithmic reliability. The final optimal model undergoes in-depth analysis via **Feature Importance** and **Residual Analysis** to interpret the prediction errors.

IV. Experiment

A. Dataset and Visualization

Visualization helps clarify data structures before moving on to machine learning training:

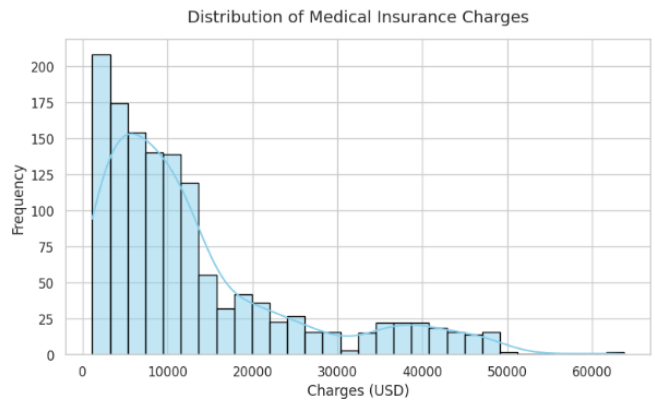

Fig. 7. Histogram - Distribution of Medical Insurance Charges

The distribution of medical costs exhibits a right-skewed pattern. The majority of observations are concentrated below USD 15,000; however, there is a notable frequency of high-cost outliers extending up to approximately USD 60,000.

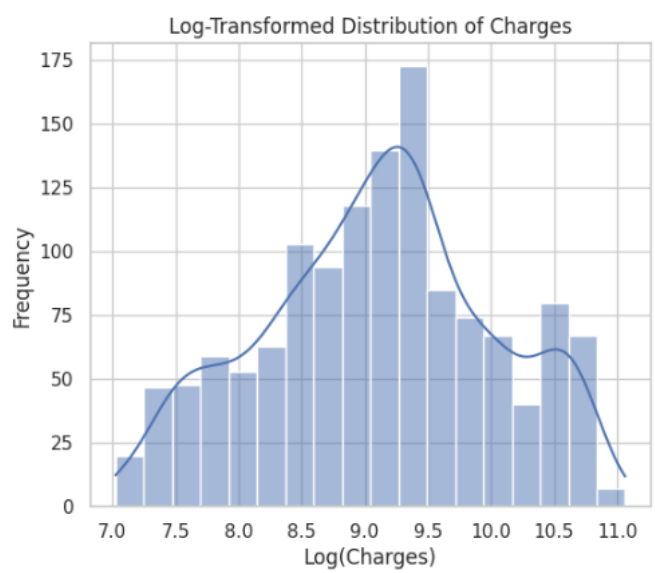

Fig. 8. Histogram - Log-Transformed Distribution of Charges

The plot illustrates the distribution after logarithmic transformation. The long tail has been compressed, resulting in a more symmetric, bell-shaped curve, which is more suitable for model fitting.

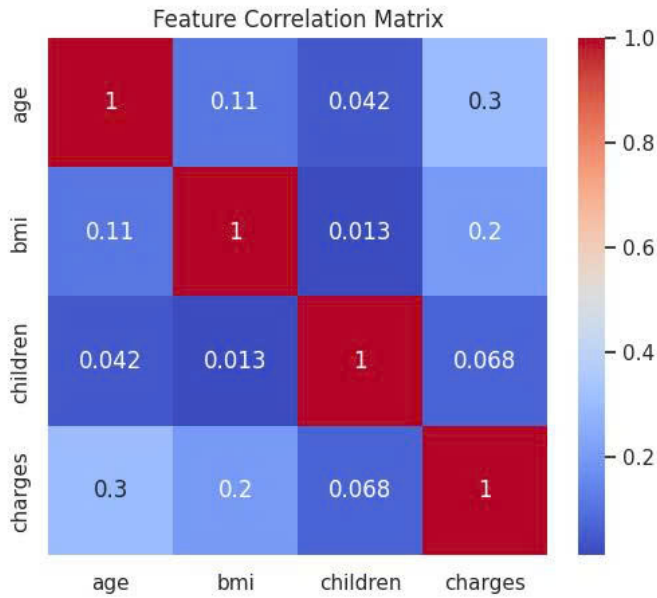

Fig. 9. Correlation Matrix Heatmap

The figure shows a correlation matrix between the variables age, bmi, children, and charges. Each value represents the linear correlation between two variables, ranging from -1 to 1. The results indicate that age and charges have a moderate positive correlation (0.3), meaning that as age increases, insurance charges tend to increase as well. BMI and charges have a weaker positive correlation (0.2), suggesting that a higher BMI may be associated with higher charges, but the relationship is not strong. Meanwhile, children and charges show a very weak correlation (0.068), indicating little to no influence. Additionally, the independent variables age, bmi, and children have very low correlations with each other, suggesting that multicollinearity among the features is minimal.

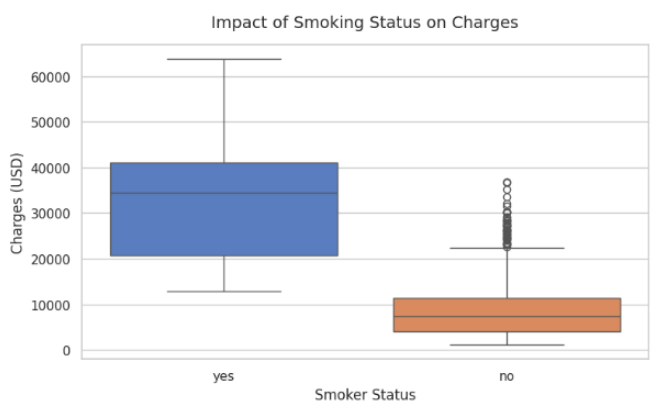

Fig. 10. Impact of Smoking Status on Charges

The plot indicates that smokers incur substantially higher and more variable medical costs compared to non-smokers. The median for smokers is markedly higher, with numerous extreme values observed, whereas non-smokers exhibit lower and more stable cost distributions.

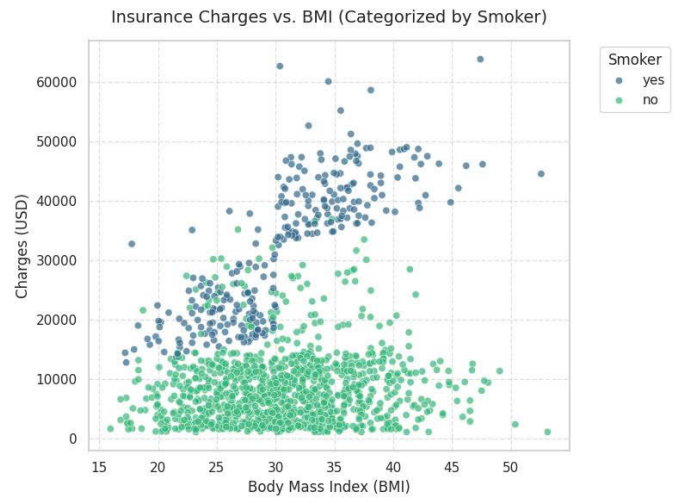

Fig. 11. Insurance Charges vs. BMI (Categorized by Smoker)

The scatter plot illustrates the relationship between medical insurance charges and Body Mass Index (BMI), stratified by smoking status. Visual analysis reveals a clear separation in baseline costs, with the smoker cohort incurring significantly higher charges compared to the non-smoker group. The most notable feature of the dataset is the presence of a strong interaction effect between these two variables. Specifically, within the non-smoking population, an increase in BMI corresponds to a relatively minor, linear marginal increase in costs. Conversely, within the smoker cohort, as BMI surpasses the clinical obesity threshold (BMI $\geq$ 30), insurance charges exhibit a dramatic, non-linear jump, shifting abruptly from the $15,000$–30,000 bracket to the $30,000$–60,000 range. This distinct distributional characteristic underscores the critical importance of feature engineering; specifically, integrating interaction terms between smoking status and the obesity threshold is essential for machine learning algorithms to fully capture the underlying data structure and optimize the predictive accuracy of subsequent regression models.

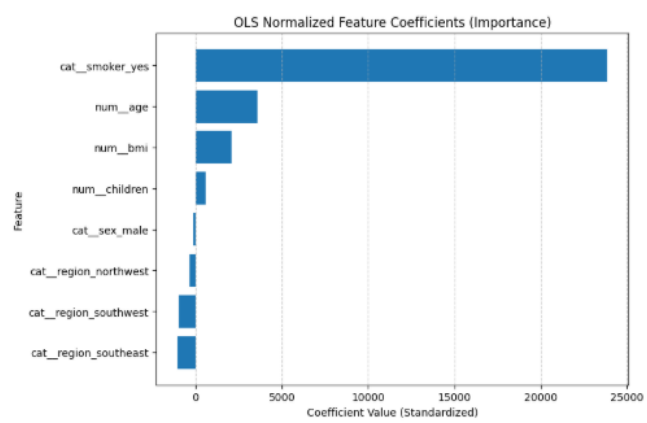

Fig. 12. OLS Normalized Feature Coefficients

Using standardized linear coefficients from the OLS model, the plot visualizes the relative influence of input features.

Empirical results indicate that smoking status (smoker_yes), age, and body mass index (bmi) are the three factors with the greatest impact on individual medical costs.

B. Result and Comparison

Experimental results demonstrate a clear improvement in overall model performance before and after applying the Log Transformation technique:

TABLE I
MODEL WITH RAW TARGET

Model	MAE	RMSE	R^2
Predict Mean	9090.99	12085.70	-0.0025
OLS	4234.98	6123.35	0.7389
Ridge	4236.15	6123.33	0.7389
Lasso	4235.44	6122.79	0.7389
SVR	3762.08	8379.23	0.5120

For the raw target, linear models—including OLS, Ridge, and Lasso—yield nearly identical performance, with an RMSE of approximately USD 6,123 and an R^2 of around 0.738. In contrast, the SVR algorithm struggles with the dispersion caused by outliers in the untransformed data, resulting in suboptimal performance.

TABLE II
MODEL WITH LOG(CHARGES))

Model	MAE	RMSE	R^2
Predict Mean	8603.16	13053.72	-0.0976
OLS	4303.15	8427.68	0.5048
Ridge	4300.52	8420.08	0.5057
Lasso	4274.81	8365.63	0.5121
SVR	2351.20	4863.63	0.8342

After applying the log transformation (log target), the SVR model shows a substantial improvement and outperforms all other approaches. According to the recorded results, SVR achieves the lowest RMSE (USD 4,863.63), with an MAE of only USD 2,351.20 and a superior R^2 of 0.834. The logarithmic scaling reduces the disparity among large values, enabling SVR (with an RBF kernel) to learn more optimal distance relationships compared to conventional linear parametric models.

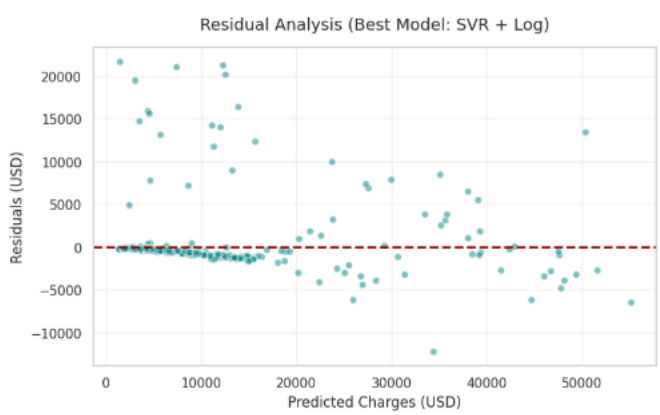

Fig. 13. Residual Analysis (Best Model: SVR + Log)

Best Model Analysis: The residual plot of the SVR model with log transformation on the independent test set shows that the errors are randomly and densely distributed around the reference line $y = 0$. No discernible patterns or systematic structures remain in the residuals, indicating that the SVR model has successfully captured the underlying relationships in the data and maintains strong generalization capability when applied to new medical cost observations.

V. DISCUSSION

The experimental results elucidate the complex characteristics of medical cost data and highlight the inherent limitations of each modeling approach:

- **Impact of Log-transformation and RMSE Sensitivity:** By utilizing the `TransformedTargetRegressor` with an inverse function ($np.expm1$), evaluation metrics (MAE, RMSE) are reverted to the original monetary unit (USD). In this original space, RMSE proves to be extremely sensitive to large errors (outliers). While log-transformation stabilizes variance—addressing *Heteroscedasticity*—and improves model fit for low-cost ranges, it does not provide a breakthrough for linear models if the noise structure is not strictly multiplicative. A few mispredictions in the "high-cost" segment are sufficient to significantly inflate the overall RMSE.
- **Data Fragmentation and Linear Constraints:** The dataset exhibits sharp boundary separation based on *smoking status*. The relationship between BMI and charges in the smoker group is non-linear or possesses a markedly different slope compared to non-smokers. Algorithms such as OLS, Ridge, and Lasso—even when combined with log-transformation—fundamentally search for linear hyperplanes and struggle to capture these complex cross-interactions.
- **Representational Power of SVR:** Overcoming the weaknesses of linear models, SVR with an RBF kernel demonstrates superior performance. By mapping data into high-dimensional spaces via the *kernel trick*, SVR can "bend" decision boundaries to flexibly capture non-linear features, such as the interaction between BMI and smoking status, thereby enhancing overall accuracy.

VI. CONCLUSION

This research has presented a systematic approach to estimating medical insurance charges based on demographic factors and lifestyle habits. Facing the challenges of heavy-tailed distributions and extreme outliers inherent in healthcare data, this study evaluated the effectiveness of Log-transformation techniques combined with a range of machine learning models, from linear approaches (OLS, Ridge, Lasso) to non-linear methods (SVR).

The empirical findings demonstrate that while Log-transformation successfully stabilizes variance, linear models exhibit clear limitations in capturing the full complexity and cross-interactions within the data—most notably the non-linear impact of BMI among smokers. In contrast, the Support Vector

Regression (SVR) model utilizing an RBF kernel proved superior due to its high-dimensional mapping capabilities, which allow for the flexible handling of non-linear decision boundaries and yield the highest predictive accuracy on the dataset. Furthermore, the results empirically reaffirm that smoking status and BMI are the two primary drivers of escalating medical costs.

Although the proposed pipeline (SVR integrated with Log-transformation) yielded promising results on the simulated dataset, real-world cost prediction necessitates further enhancements. Transitioning toward model architectures capable of automatically learning interaction terms—such as tree-based ensembles or Neural Networks—combined with the inclusion of authentic clinical data and Explainable AI (XAI) tools, will be essential steps in moving these models from theoretical frameworks to practical applications within modern healthcare financial systems.

VII. Limitations and Future Work

A. Objective Limitations

- **Absence of Interaction Terms:** The current pipeline primarily utilizes independent raw features. The under-performance of linear models reflects a lack of manually engineered interaction terms (e.g., $BMI \times smoker$) which would help the model understand the joint impact of these factors.
- **Dataset Constraints:** The utilized Kaggle dataset is oriented towards simulation and lacks high-weight clinical variables (e.g., pre-existing conditions, detailed smoking intensity). Consequently, errors in the high-cost group remain unpredictable as the models lack sufficient data to explain extreme price spikes.

B. Future Research Directions

- **Advanced Modeling Approaches:** Future work should transition from forcing linear models to handle non-linearities toward *tree-based models* (e.g., Random Forest, XGBoost) or Neural Networks (MLP). These architectures inherently capture cross-interactions and handle complex data branching without requiring rigorous log-transformations.
- **Enhanced Feature Engineering:** Implementing a dedicated feature design step to create new variables combining BMI, age, and smoking status to empower baseline machine learning models.
- **Practical Application with Transparency:** Developing a Cost Estimator integrated with **Local Explainability** (e.g., using SHAP values). Such a system would explicitly show users how much their costs are influenced by BMI or smoking, accompanied by a disclaimer regarding the intrinsic limits of simulated data.

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